Iteration or Loops in Python

10 October 2024 09:23

'while' loop in Python

The while loop in python is used to repeatedly execute a block of code as long as a given condition is true. It is a fundamental control flow statement that allows for. the creation of loop that can iterate for indefinite number of times depending upon the condition.

Syntax of the 'while' loop

```
while condition:
    #code block to be executed
```

- **'condition':** This is a boolean expression that evaluates to 'True' or 'False'. The loop continues as long as this condition remains 'True'.
- **'code-block':** The block of code that you want to execute repeatedly. It must be indented consistently.

Multiplication Table

1 * v = v
2 * v = 2v
3 x v = 3v
⋮
so on.

If the value of v = 5

fixed

1 * 5 = 5
2 * 5 = 10
3 * 5 = 15
⋮

input output

n * v = n.v

starts from 1 and goes upto 't'

input

```
v = int(input('Enter any positive integer: '))
t = int(input("How many times?"))
n = 1
while n <= t:
    p = n * v
    print(n,'*',v,'=',p)
    n += 1  # n = n + 1
```

V = 5
t = 5
n = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6
p = ~~5~~ ~~10~~ ~~15~~ ~~20~~ 25

1 * 5 = 5
2 * 5 = 10
3 * 5 = 15
4 * 5 = 20
5 * 5 = 25

Infinite Loop:

A while loop can potentially become an infinite loop if the condition always evaluates to true. This usually happens if the condition is never updated within the loop.

Ctrl + C to terminate the program.

Keyboardinterrupt.

```
while True:
    # block of code
```

For example:

```python
c = 0
while True:
    print('I am an infinite loop')
    c += 1
    print(c)
```

Output:

```
OUTPUT   PROBLEMS   PORTS   TERMINAL   ···   Python

I am an infinite loop
35877
I am an infinite loop
35878
I am an infinite loop
35879
I am an infinite loop
35880
I am an infinite loop
35881
I am an infinite loop
35882
Traceback (most recent call last):
  File "c:\Users\bluej\Documents\Students\JiyaGaurDeensAcadem
y\Coding\chapter1\three.py", line 6, in <module>
    print(c)
KeyboardInterrupt
PS C:\Users\bluej\Documents\Students\JiyaGaurDeensAcademy\Coc
ing>
```

To avoid an infinite loop. Ensure that condition will eventually become false. You can use 'break' to exit the loop prematurely.

- **'break':** Exits the current loop entirely.

For example:

```python
v = int(input('Enter any positive integer: '))
t = int(input("How many times?"))
n = 1
while True:
    p = n * v
    print(n,'*',v,'=',p)
    if n == t:
        break #terminate the loop

    n += 1  # n = n + 1
```

Output:

```
Enter any positive integer: 987654321
How many times?10
1 * 987654321 = 987654321
2 * 987654321 = 1975308642
3 * 987654321 = 2962962963
4 * 987654321 = 3950617284
5 * 987654321 = 4938271605
6 * 987654321 = 5925925926
7 * 987654321 = 6913580247
8 * 987654321 = 7901234568
9 * 987654321 = 8888888889
10 * 987654321 = 9876543210
```

2) 1 (0
-0
1

remainder

- **'continue':** Skip the rest of the code inside the loop for the current iteration and moves to the next iteration.

- **'continue'**: Skip the rest of the code inside the loop for the current iteration and moves to the next iteration.

```
n = int(input('enter number of terms'))
while n > 0:
    if n % 2 == 0:
        n -= 1
        continue

    print('n =', n)
    n -= 1
```

n = 10 9 8 7 6 5 4 3 2 1 0

n = 9
n = 7
n = 5
n = 3
n = 1

WAP to input number of terms and print the series:

i. 1, 4, 9, 16,…
   Input: Enter number of terms: 7
   Output: 1, 4, 9, 16, 25, 36, 49

ii. 1, 2, 4, 7, 11, 16, 22,…
   Input: Enter number of terms: 7
   Output: 1, 2, 4, 7, 11, 16, 22

```
terms=int(input("Enter the number of terms: "))
start=1
p=1
while start<terms:
    print(p,end=', ')
    p=p+start
    start+=1
p=p+start
print(p)
```

terms = 5
start = 1 2 3 4 5
p = 1 2 4 7 11

1, 2, 4, 7, 11

Home work:

iii. 3, 6, 9, 12,…
iv. 4, 8, 16, 32,…
v. 1.5, 3.0, 4.5, 6.0,…
vi. 0, 7, 26,…
vii. 1, 9, 25, 49,…
viii. 0, 3, 8, 15,…
ix. 2, 5, 10, 17,…
x. 24, 99, 224, 399,…

2. Write a program to input any 10 numbers (including positive and negative) Perform the following tasks:
(a) Count and display the positive numbers
(b) Count and display the negative numbers
(c) Display the sum of positive numbers
(d) Display the sum of negative numbers

3. Write a program to display all the numbers between m and n input from the keyboard (where m>0, n>0), check and display the numbers that are perfect square. e.g. 25, 36, 49, are said to be perfect square numbers. [A number is said to be a perfect square if its square root is an integer number.]

(where m>0, n>0), check and display the numbers that are perfect square. e.g. 25, 36, 49, are said to be perfect square numbers. [A number is said to be a perfect square if its square root is an integer number.]

m = 10 ← lower bound
n = 50 ← upper bound.

while m <= n:
m += 1

m = 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22 23, 24, 25 n = 50
26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36

√10 = 3.1
√11 = 3.2
√12 = 3.3
√13 = 3.6
√14 = 3.8
√15 = 3.9
√16 = 4.0 ← perfect square. 16 is selected for printing.

√25 = 5.0 ← " " 25 " " "

3.6
3 √13
3 9
36 400
396
4

while m <= n:
rt = int(m ** 0.5)
if rt**2 == m:
print(f"{m} is a perfect square");
m += 1

√12 = (3.3) int
rt = 3
3 X 3 == m ← false

√16 = 4.0
rt = 4
4X4 == m true.

Home work

1. Write a program to display all the 'Buzz Numbers' between p and q (where p<q). A 'Buzz Number' is the number which ends with 7 or divisible by 7.
2. Write a program to input marks in English, Maths and Science of 10 students who have passed ICSE Examination 2014. Now, perform the following tasks:
   (a) Find and display the number of students, who have secured 95% or more in aggregate.
   (b) Find and display the number of students, who have secured 90% or more in English, Maths and Science.

3. Write a program in Python to input number of terms say N.
   Print the series and display the sum of the following series:
   a. 1 + 4 + 9 + …
   b. 1 + 1/2 + 1/3 + … +4
   c. 1/2 + 2/3 + 3/4 +…
   d. 2 - 4 + 6 - 8 + 10 - …
   e. (1*2) + (2*3) + (3*4) + …

seven_div = ~~0~~ + 3
end_with_7 = ~~4~~ ~~5~~ 7
p = 10 q = 20.
~~11~~ ~~12~~ ~~13~~ ~~14~~ ~~15~~ ~~16~~ ~~17~~ 18 19 20 21

```
4
5  p=int(input("Enter lower bound: "))
```

```
p=int(input("Enter lower bound: "))
q=int(input("Enter upper bound: "))
num=p
if (p<=0 or q<=0) or p>=q:
    print("Invalid input")
    exit(0)
output=""
while num<=q:
    seven_div=num%7
    end_with_7=num%10
    if seven_div==0 or end_with_7==7:
        output+=str(num)+","
    num+=1
print(output[0:output.rindex(",")])
```

end_with_7 = ~~14~~ ~~15~~ 7

p = 10 q = 20

num = ~~10~~ ~~11~~ ~~12~~ ~~13~~ ~~14~~ ~~15~~ ~~16~~ ~~17~~ 18 19, 20, 21

output = '7, 17,'

[7, 1 7,]

0 1 2 3 4 R

[0 : 4]

4 - 1 = 3

More questions for home work:

Q4. Write a program to input a number and count the number of digits the. program further checks whether the number contained odd number of digits or even number of digits and display the output accordingly. For example:
Enter a number: 194
Number of digits = 3
The number contains odd number of digits.

Q5. Write a program to input a number and display the new number after reversing the digits of the original number. The program also display the absolute difference between the original number and the reverse number. For example,:
Enter a number: 194
Reverse: 491
Absolute difference: abs(194 - 491) = 297
[Hint: use abs() method of Python]

Q6. Write a program in python to find and display the sum of the following series, up to N terms.

(i) $S = a^2 + \frac{a^2}{2} + \frac{a^2}{3} + \cdots + \frac{a^2}{N}$

(ii) $S = a + \frac{a^2}{2} + \frac{a^3}{3} + \cdots + \frac{a^N}{N}$

(iii) $S = \frac{1^2}{a} + \frac{3^2}{a^2} + \frac{5^2}{a^3} + \frac{7^2}{a^4} + \cdots + \frac{(B \times 2 - 1)^2}{a^N}$